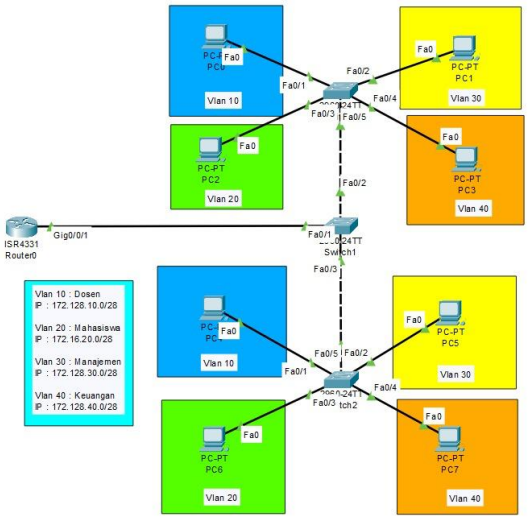
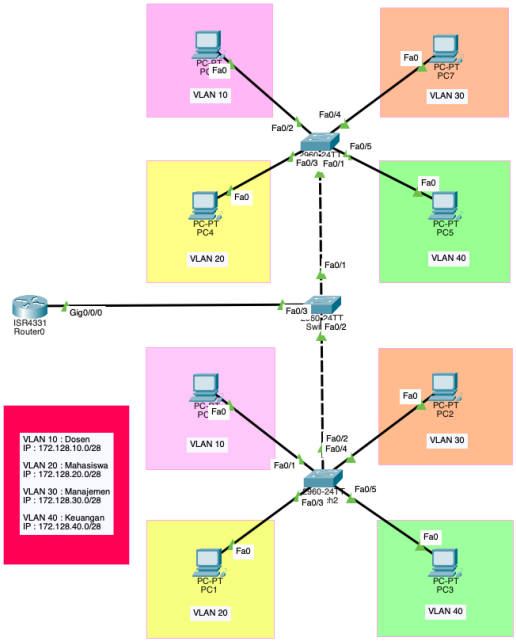
1. Jelaskan kapan InterVLAN Routing di terapkan!

   InterVLAN Routing diterapkan saat kita memiliki beberapa VLAN yang berbeda (subnet dan IP berbeda) yang perlu berkomunikasi satu sama lain, karena secara *default*, VLAN terisolasi di Layer 2 dan harus dilewatkan melalui perangkat Layer 3 (Router atau Layer 3 Switch) untuk pertukaran paket (berkomunikasi).

2. Buatlah rancangan topologi sesuai instruksi pada gambar di bawah





Switch 1:

```
Switch>enable
Switch#conf  t
Switch(config)#hostname  lab
lab(config)#vlan  10
lab(config-vlan)#name  Dosen
lab(config-vlan)#int  fa0/2
lab(config-if)#sw  mo  ac
lab(config-if)#sw  ac  vlan  10
lab(config-if)#exit
lab(config)#vlan  20
lab(config-vlan)#name  Mahasiswa
lab(config-vlan)#int  fa0/3
lab(config-if)#sw  mo  ac
lab(config-if)#sw  ac  vlan 20
lab(config-if)#exit
lab(config)#vlan  30
lab(config-vlan)#name  Manajemen
lab(config-vlan)#int  fa0/4
lab(config-if)#sw  mo  ac
lab(config-if)#sw  ac  vlan  30
lab(config-if)#exit
lab(config)#vlan  40
lab(config-vlan)#name  Keuangan
lab(config-vlan)#int  fa0/5
lab(config-if)#sw  mo  ac
lab(config-if)#sw  ac  vlan  40
lab(config-if)#do  show  vlan
```

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   Dosen                            active    Fa0/2
20   Mahasiswa                        active    Fa0/3
30   Manajemen                        active    Fa0/4
40   Keuangan                         active    Fa0/5
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
10   enet  100010     1500  -      -      -        -    -        0      0
```

Switch 2:

Switch>enable

Switch#conf t

Switch(config)#hostname fikom

fikom(config)#vlan 10

fikom(config-vlan)#Name Dosen

fikom(config-vlan)#vlan 20

fikom(config-vlan)#Name Mahasiswa

fikom(config-vlan)#vlan 30

fikom(config-vlan)#name Manajemen

fikom(config-vlan)#vlan 40

fikom(config-vlan)#name Keuangan

fikom(config-vlan)#do show vlan

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   Dosen                            active
20   Mahasiswa                        active
30   Manajemen                        active
40   Keuangan                         active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
```

fikom(config-vlan)#exit

fikom(config)#int range fa0/1-3

fikom(config-if-range)#sw mo tr

```
fikom(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
```

Router:

Router>enable

Router#conf t

Router(config)#int gig0/0/0

Router(config-if)#no shutdown

Router(config-if)#int gig0/0/0

Router(config-if)#int gig0/0/0.10

Router(config-subif)#encapsulation dot1q 10

Router(config-subif)#ip address 172.128.10.1 255.255.255.240

Router(config-subif)#exit

Router(config)#int gig0/0/0.20

Router(config-subif)#encapsulation dot1q 20

Router(config-subif)#ip address 172.128.20.1 255.255.255.240

Router(config-subif)#exit

Router(config-subif)#int gig0/0/0.30

Router(config-subif)#encapsulation dot1q 30

Router(config-subif)#ip address 172.128.30.1 255.255.255.240

Router(config-subif)#exit

Router(config)#int gig0/0/0.40

Router(config-subif)#encapsulation dot1q 40

Router(config-subif)#ip address 172.128.40.1 255.255.255.240

Router(config-subif)#exit

Router(config)exit

Router#show arp

```
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  172.128.10.2            14  0005.5EEC.18DA  ARPA   GigabitEthernet0/0/0.10
Internet  172.128.10.3            10  0005.5E95.AA36  ARPA   GigabitEthernet0/0/0.10
Internet  172.128.10.4            7   0005.5EEC.18DA  ARPA   GigabitEthernet0/0/0.10
Internet  172.128.20.2            14  000C.CF2A.5567  ARPA   GigabitEthernet0/0/0.20
Internet  172.128.20.3            4   0009.7C59.6E5C  ARPA   GigabitEthernet0/0/0.20
Internet  172.128.30.2            11  0001.6450.0366  ARPA   GigabitEthernet0/0/0.30
Internet  172.128.30.3            14  0001.64C1.5430  ARPA   GigabitEthernet0/0/0.30
Internet  172.128.40.2            14  00D0.587C.6725  ARPA   GigabitEthernet0/0/0.40
Internet  172.128.40.3            11  0006.2A6B.8B20  ARPA   GigabitEthernet0/0/0.40
```

Switch 3:

Switch>enable

Switch#conf t

Switch(config)#hostname umi

umi(config)#vlan 10

umi(config-vlan)#name Dosen

umi(config-vlan)#int fa0/1

umi(config-if)#sw mo ac

umi(config-if)#sw ac vlan 10

umi(config-if)#exit

umi(config)#vlan 20

umi(config-vlan)#name Mahasiswa

umi(config-vlan)#int fa0/3

umi(config-if)#sw mo ac

umi(config-if)#sw ac vlan 20

umi(config-if)#exit

umi(config)#vlan 30

umi(config-vlan)#name Manajemen

umi(config-vlan)#int fa0/4

umi(config-if)#sw mo ac

umi(config-if)#sw ac vlan 30

umi(config-if)#exit

umi(config)#vlan 40

umi(config-vlan)#name Keuangan

umi(config-vlan)#int fa0/5

umi(config-if)#sw mo ac

umi(config-if)#sw ac vlan 40

umi(config-if)#do show vlan

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   Dosen                            active    Fa0/1
20   Mahasiswa                        active    Fa0/3
30   Manajemen                        active    Fa0/4
40   Keuangan                         active    Fa0/5
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
10   enet  100010     1500  -      -      -        -    -        0      0
```

Ping Pengujian:

a) Vlan yang sama diswitch yang berbeda

Ping PC0 ke PC6 (VLAN 10)

c:\>ping 172.128.10.3

```
Pinging 172.128.10.3 with 32 bytes of data:

Reply from 172.128.10.3: bytes=32 time<1ms TTL=128
Reply from 172.128.10.3: bytes=32 time=2ms TTL=128
Reply from 172.128.10.3: bytes=32 time=2ms TTL=128
Reply from 172.128.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 172.128.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 1ms
```

Ping PC4 ke PC1 (VLAN 20)

c:\>ping 172.128.20.3

```
Pinging 172.128.20.3 with 32 bytes of data:

Reply from 172.128.20.3: bytes=32 time<1ms TTL=128
Reply from 172.128.20.3: bytes=32 time<1ms TTL=128
Reply from 172.128.20.3: bytes=32 time<1ms TTL=128
Reply from 172.128.20.3: bytes=32 time<1ms TTL=128

Ping statistics for 172.128.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping PC7 ke PC2 (VLAN 30)

c:\>ping 172.128.30.2

```
Pinging 172.128.30.2 with 32 bytes of data:

Reply from 172.128.30.2: bytes=32 time<1ms TTL=128
Reply from 172.128.30.2: bytes=32 time<1ms TTL=128
Reply from 172.128.30.2: bytes=32 time<1ms TTL=128
Reply from 172.128.30.2: bytes=32 time=6ms TTL=128

Ping statistics for 172.128.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 6ms, Average = 1ms
```

Ping PC5 ke PC3 (VLAN 40)

c:\>ping 172.128.40.3

```
Pinging 172.128.40.3 with 32 bytes of data:

Reply from 172.128.40.3: bytes=32 time<1ms TTL=128
Reply from 172.128.40.3: bytes=32 time<1ms TTL=128
Reply from 172.128.40.3: bytes=32 time<1ms TTL=128
Reply from 172.128.40.3: bytes=32 time<1ms TTL=128

Ping statistics for 172.128.40.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

b) Vlan yang berbeda di switch yang sama

Ping PC6 ke PC4 (switch 1)

c:\>ping 172.128.20.2

```
Pinging 172.128.20.2 with 32 bytes of data:

Reply from 172.128.20.2: bytes=32 time<1ms TTL=127
Reply from 172.128.20.2: bytes=32 time<1ms TTL=127
Reply from 172.128.20.2: bytes=32 time=1ms TTL=127
Reply from 172.128.20.2: bytes=32 time<1ms TTL=127

Ping statistics for 172.128.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Ping PC4 ke PC7 (swicth 1)

c:\>ping 172.128.30.3

```
Pinging 172.128.30.3 with 32 bytes of data:

Reply from 172.128.30.3: bytes=32 time=1ms TTL=127
Reply from 172.128.30.3: bytes=32 time<1ms TTL=127
Reply from 172.128.30.3: bytes=32 time<1ms TTL=127
Reply from 172.128.30.3: bytes=32 time<1ms TTL=127

Ping statistics for 172.128.30.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Ping PC7 ke PC5 (Switch 1)

c:\>ping 172.128.40.2

```
inging 172.128.40.2 with 32 bytes of data:

eply from 172.128.40.2: bytes=32 time<1ms TTL=127
eply from 172.128.40.2: bytes=32 time<1ms TTL=127
eply from 172.128.40.2: bytes=32 time<1ms TTL=127
eply from 172.128.40.2: bytes=32 time<1ms TTL=127

ing statistics for 172.128.40.2:
   Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
pproximate round trip times in milli-seconds:
   Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping PC0 ke PC1 (Switch 3)

c:\>ping 172.128.20.3

```
Pinging 172.128.20.3 with 32 bytes of data:

Reply from 172.128.20.3: bytes=32 time=12ms TTL=127
Reply from 172.128.20.3: bytes=32 time<1ms TTL=127
Reply from 172.128.20.3: bytes=32 time<1ms TTL=127
Reply from 172.128.20.3: bytes=32 time=1ms TTL=127

Ping statistics for 172.128.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 3ms
```

Ping PC1 ke PC2 (Switch 3)

c:\>ping 172.128.30.2

```
Pinging 172.128.30.2 with 32 bytes of data:

Reply from 172.128.30.2: bytes=32 time<1ms TTL=127
Reply from 172.128.30.2: bytes=32 time<1ms TTL=127
Reply from 172.128.30.2: bytes=32 time=36ms TTL=127
Reply from 172.128.30.2: bytes=32 time<1ms TTL=127

Ping statistics for 172.128.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 36ms, Average = 9ms
```

Ping PC2 ke PC3 (Switch 3)

c:\>ping 172.128.40.3

```
Pinging 172.128.40.3 with 32 bytes of data:

Reply from 172.128.40.3: bytes=32 time=1ms TTL=127
Reply from 172.128.40.3: bytes=32 time<1ms TTL=127
Reply from 172.128.40.3: bytes=32 time<1ms TTL=127
Reply from 172.128.40.3: bytes=32 time=14ms TTL=127

Ping statistics for 172.128.40.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 14ms, Average = 3ms
```

c) Vlan yang berbeda di switch yang berbeda

Ping PC6 ke PC1

c:\>ping 172.128.20.3

```
Pinging 172.128.20.3 with 32 bytes of data:

Reply from 172.128.20.3: bytes=32 time<1ms TTL=127
Reply from 172.128.20.3: bytes=32 time<1ms TTL=127
Reply from 172.128.20.3: bytes=32 time<1ms TTL=127
Reply from 172.128.20.3: bytes=32 time<1ms TTL=127

Ping statistics for 172.128.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping PC4 ke PC0

c:\>ping 172.128.10.4

```
Pinging 172.128.10.4 with 32 bytes of data:

Reply from 172.128.10.4: bytes=32 time<1ms TTL=127
Reply from 172.128.10.4: bytes=32 time<1ms TTL=127
Reply from 172.128.10.4: bytes=32 time=1ms TTL=127
Reply from 172.128.10.4: bytes=32 time=2ms TTL=127

Ping statistics for 172.128.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

Ping PC7 ke PC3

c:\>ping 172.128.40.3

```
Pinging 172.128.40.3 with 32 bytes of data:

Reply from 172.128.40.3: bytes=32 time<1ms TTL=127
Reply from 172.128.40.3: bytes=32 time<1ms TTL=127
Reply from 172.128.40.3: bytes=32 time=19ms TTL=127
Reply from 172.128.40.3: bytes=32 time<1ms TTL=127

Ping statistics for 172.128.40.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 19ms, Average = 4ms
```

Ping PC 5 ke PC2

c:\>ping 172.128.30.2

```
Pinging 172.128.30.2 with 32 bytes of data:

Reply from 172.128.30.2: bytes=32 time=15ms TTL=127
Reply from 172.128.30.2: bytes=32 time=1ms TTL=127
Reply from 172.128.30.2: bytes=32 time<1ms TTL=127
Reply from 172.128.30.2: bytes=32 time<1ms TTL=127

Ping statistics for 172.128.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 15ms, Average = 4ms
```